

Assignment 01 Solution

Solutions

Solution to Problem 1

Given Information

The sample space is

$$S = \{1, 2, 3, 4, 5, 6, 7, 8\}.$$

The events are defined as:

$$A = \{\text{even numbers}\}, \quad B = \{\text{numbers greater than 3}\}, \quad C = \{\text{numbers divisible by 3}\}.$$

(a) Explicit Description of Events

Listing elements of each event from the sample space:

$$A = \{2, 4, 6, 8\}, \quad B = \{4, 5, 6, 7, 8\}, \quad C = \{3, 6\}.$$

(b) High-Risk Requests

A request is classified as high-risk if it satisfies at least two of the three conditions A , B , and C .

This occurs when the request belongs to any pairwise intersection:

$$(A \cap B) \cup (A \cap C) \cup (B \cap C).$$

We compute each intersection:

$$A \cap B = \{4, 6, 8\},$$

$$A \cap C = \{6\},$$

$$B \cap C = \{6\}.$$

Thus, the set of high-risk requests is

$$(A \cap B) \cup (A \cap C) \cup (B \cap C) = \{4, 6, 8\}.$$

(c) Low-Risk Requests

A request is classified as low-risk if it satisfies neither condition A nor condition C .

This corresponds to the complement of $A \cup C$:

$$(A \cup C)^C = A^C \cap C^C.$$

We compute:

$$A^C = \{1, 3, 5, 7\}, \quad C^C = \{1, 2, 4, 5, 7, 8\}.$$

Therefore,

$$A^C \cap C^C = \{1, 5, 7\}.$$

(d) Justification of the High-Risk Expression

To satisfy at least two of the three conditions A , B , and C , a request must satisfy one of the following:

- both A and B ,
- both A and C , or
- both B and C .

These cases are mutually inclusive and collectively exhaustive for “at least two” conditions. Therefore, the set of all high-risk requests is correctly expressed as

$$(A \cap B) \cup (A \cap C) \cup (B \cap C).$$

(e) Verification of the Set Identity

We examine the identity:

$$[(A \cap B) \cup C]^C \stackrel{?}{=} (A^C \cup B^C) \cap C^C.$$

Using De Morgan’s laws,

$$[(A \cap B) \cup C]^C = (A \cap B)^C \cap C^C.$$

Applying De Morgan’s law again,

$$(A \cap B)^C = A^C \cup B^C.$$

Thus,

$$[(A \cap B) \cup C]^C = (A^C \cup B^C) \cap C^C.$$

Hence, the identity holds.

$$\boxed{[(A \cap B) \cup C]^C = (A^C \cup B^C) \cap C^C}$$

Solution to Problem 2

(a) Experiment

The experiment consists of rolling the given biased six-sided die once and observing the number that appears on the top face.

(b) Sample Space

The sample space consists of all possible outcomes when the die is rolled:

$$S = \{1, 2, 3, 4, 5, 6\}.$$

(c) Events Mentioned

The colleague refers to the following events:

$$E = \{\text{even outcomes}\} = \{2, 4, 6\}$$

$$O = \{\text{odd outcomes}\} = \{1, 3, 5\}$$

$$G = \{\text{outcomes greater than 3}\} = \{4, 5, 6\}$$

$$N = \{\text{outcomes not greater than 3}\} = \{1, 2, 3\}$$

(d) Disjoint Events

Two events are disjoint if their intersection is the empty set.

$$E \cap O = \emptyset$$

$$G \cap N = \emptyset$$

Thus, events E and O are disjoint, and events G and N are also disjoint.

(e) Additive Axiom

The Additive Axiom of Probability applies to disjoint events. Therefore, it applies to:

$$P(E \cup O) = P(E) + P(O)$$

and to:

$$P(G \cup N) = P(G) + P(N)$$

(f) Probabilities of Even and Odd Faces

Let the probability of each odd face be x . Since each even face is twice as likely, each even face has probability $2x$.

There are three odd faces and three even faces, so:

$$3x + 3(2x) = 1$$

$$3x + 6x = 1$$

$$9x = 1$$

$$x = \frac{1}{9}$$

Thus:

$$P(O) = 3x = \frac{3}{9} = \frac{1}{3}$$

$$P(E) = 3(2x) = \frac{6}{9} = \frac{2}{3}$$

(g) Probability of Rolling More Than 3

$$G = \{4, 5, 6\}, \quad N = \{1, 2, 3\}$$

$$P(G) = P(4) + P(5) + P(6)$$

$$= \frac{2}{9} + \frac{1}{9} + \frac{2}{9} = \frac{5}{9}$$

$$P(N) = P(1) + P(2) + P(3)$$

$$= \frac{1}{9} + \frac{2}{9} + \frac{1}{9} = \frac{4}{9}$$

Since

$$P(G) > P(N),$$

the die is more likely to roll a number greater than 3.

Conclusion

The probability of rolling a number greater than 3 is $\frac{5}{9}$, which is greater than the probability of not rolling a number greater than 3. Therefore, the colleague's question is answered affirmatively.

Solution 3

We can use the Venn diagram in Figure 1 to better visualize the events in this problem. Assume

$$P(A) = a, \quad P(B) = b, \quad P(C) = c.$$

Note that the assumptions about independence and disjointness of sets are already represented in the figure.

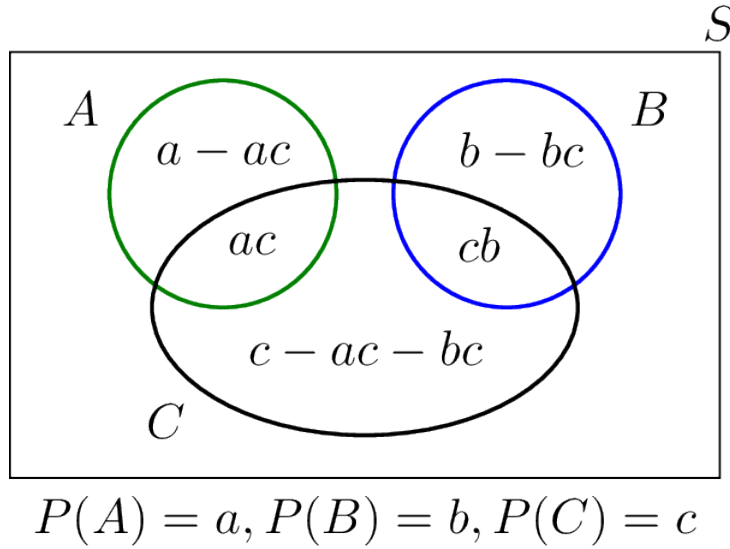


Figure 1: Venn diagram for Problem 3

Using the properties of unions and the given information, we have:

$$\begin{aligned} P(A \cup C) &= a + c - ac = \frac{20}{30}, \\ P(B \cup C) &= b + c - bc = \frac{30}{40}, \\ P(A \cup B \cup C) &= a + b + c - ac - bc = \frac{11}{12}. \end{aligned}$$

By subtracting the third equation from the sum of the first two equations, we immediately obtain

$$c = \frac{1}{2},$$

which then gives

$$a = \frac{1}{3} \quad \text{and} \quad b = \frac{1}{2}.$$

Solution 4

(a) Find U , I , and D

Union U :

$$A_2 = [0, 1/2), \quad A_3 = [0, 2/3), \quad A_4 = [0, 3/4), \dots$$

The upper bounds approach 1 as $n \rightarrow \infty$, so:

$$U = \bigcup_{n=2}^{\infty} A_n = [0, 1)$$

Intersection I :

$$B_2 = [1/4, 3/4], \quad B_3 = [1/9, 8/9], \quad B_4 = [1/16, 15/16], \dots$$

The lower bounds increase to 0, the upper bounds decrease to 1, so choosing least common area:

$$I = \bigcap_{n=2}^{\infty} B_n = \left[\frac{1}{4}, \frac{3}{4} \right)$$

Union D :

$$C_2 = (-1/2, 1/2), \quad C_3 = (-1/3, 1/3), \quad C_4 = (-1/4, 1/4), \dots$$

The union covers all numbers between the smallest and largest intervals:

$$D = \bigcup_{n=2}^{\infty} C_n = (-1/2, 1/2)$$

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(b) Find $U \cap I$ and $U \cup D$

$$U \cap I = [0, 1) \cap \left[\frac{1}{4}, \frac{3}{4} \right) = \left[\frac{1}{4}, \frac{3}{4} \right)$$

$$U \cup D = [0, 1) \cup (-1/2, 1/2) = (-1/2, 1)$$

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(c) Check $E = \bigcap_{n=2}^{\infty} A_n \cap \bigcup_{n=2}^{\infty} C_n$

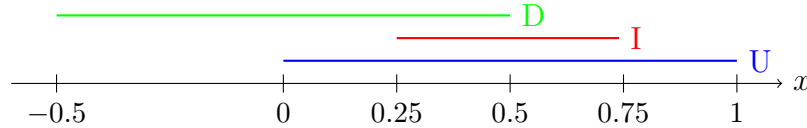
$\bigcap_{n=2}^{\infty} A_n = A_2 \cap A_3 \cap \dots = [0, 1/2)$ (smallest interval dominates).

$\bigcup_{n=2}^{\infty} C_n = (-1/2, 1/2)$

$\implies E = [0, 1/2) \cap (-1/2, 1/2) = [0, 1/2) \neq \emptyset$

$$\boxed{E = [0, 1/2)}$$

(d) Number line sketch (conceptual)



Final Answers:

$$U = [0, 1), \quad I = \left[\frac{1}{4}, \frac{3}{4} \right), \quad D = (-1/2, 1/2),$$

$$U \cap I = \left[\frac{1}{4}, \frac{3}{4} \right), \quad U \cup D = (-1/2, 1), \quad E = [0, 1/2)$$

Problem 5 Monty Hall Problem

Before any reveal, the GPU is equally likely to be behind any one of the three servers:

$$\mathbb{P}(A) = \mathbb{P}(B) = \mathbb{P}(C) = \frac{1}{3}.$$

You initially pick A . The admin (who knows all contents) opens *one* of $\{B, C\}$ that is guaranteed to be corrupted. This action does not change the probability that your original choice A was correct; it remains $1/3$. Therefore, the entire remaining probability mass $2/3$ gets concentrated on the *single* unopened server that is not A . Hence switching wins with probability $2/3$.

Case Analysis (Exhaustive). Let the true GPU location be uniformly at random among $\{A, B, C\}$. You pick A .

1. GPU at A (prob. $1/3$): The admin opens one corrupted among $\{B, C\}$. If you switch, you lose.
2. GPU at B (prob. $1/3$): The admin must open C . If you switch (to B), you win.
3. GPU at C (prob. $1/3$): The admin must open B . If you switch (to C), you win.

Thus, switching wins in 2 of the 3 equally likely worlds $\Rightarrow 2/3$.

Bayes/Conditional Probability Derivation. Let G_X denote “GPU behind server X ”, and let R_C denote the event that the admin opens server C (revealing corruption). Assume the admin *always* opens a corrupted server and, when indifferent (GPU behind A , both B and C corrupted), chooses uniformly between them. We want $\mathbb{P}(G_B | R_C)$.

$$\mathbb{P}(G_B | R_C) = \frac{\mathbb{P}(R_C | G_B) \mathbb{P}(G_B)}{\mathbb{P}(R_C)}.$$

Compute each term:

$$\mathbb{P}(G_B) = \frac{1}{3}, \quad \mathbb{P}(G_C) = \frac{1}{3}, \quad \mathbb{P}(G_A) = \frac{1}{3}.$$

$$\mathbb{P}(R_C | G_B) = 1 \quad (\text{admin must open } C),$$

$$\mathbb{P}(R_C | G_C) = 0 \quad (\text{admin cannot open the GPU door}),$$

$$\mathbb{P}(R_C | G_A) = \frac{1}{2} \quad (\text{admin chooses uniformly between } B \text{ and } C).$$

Hence

$$\mathbb{P}(R_C) = \sum_{X \in \{A, B, C\}} \mathbb{P}(R_C | G_X) \mathbb{P}(G_X) = \frac{1}{2} \cdot \frac{1}{3} + 1 \cdot \frac{1}{3} + 0 \cdot \frac{1}{3} = \frac{1}{6} + \frac{1}{3} = \frac{1}{2}.$$

Therefore

$$\mathbb{P}(G_B | R_C) = \frac{1 \cdot (1/3)}{1/2} = \frac{2}{3}.$$

So, after the admin opens a corrupted server C , the remaining closed server B contains the GPU with probability $2/3$, and switching is optimal.

Problem 6: Using a Biased Coin to Make an Unbiased Decision.

Flip the coin twice. If the outcome is heads–tails, choose the opera. If the outcome is tails–heads, choose the movies. Otherwise, repeat the process until a decision can be made.

Let A_k be the event that a decision was made at the k th round. Conditional on the event A_k , the two choices are equally likely, and we have

$$P(\text{opera}) = \sum_{k=1}^{\infty} P(\text{opera} | A_k) P(A_k) = \sum_{k=1}^{\infty} \frac{1}{2} P(A_k) = \frac{1}{2}.$$

We have used here the property

$$\sum_{k=1}^{\infty} P(A_k) = 1,$$

which is true as long as $P(\text{heads}) > 0$ and $P(\text{tails}) > 0$.

Solution 7: Counting Valid Curricula

Given: 8 lower-level courses $\{L_1, \dots, L_8\}$, 10 higher-level courses $\{H_1, \dots, H_{10}\}$. A curriculum consists of 4 lower-level and 3 higher-level courses.

(a) **No prerequisites:**

Number of ways to choose 4 lower-level courses: $\binom{8}{4}$. Number of ways to choose 3 higher-level courses: $\binom{10}{3}$.

Therefore, total curricula:

$$\binom{8}{4} \cdot \binom{10}{3} = 70 \cdot 120 = 8400.$$

(b) **With prerequisites:**

Let us consider the constraints:

- If a curriculum includes a course from $\{H_1, \dots, H_5\}$, it must include L_1 .
- If a curriculum includes a course from $\{H_6, \dots, H_{10}\}$, it must include L_2 and L_3 .

Let us split the higher-level courses:

- (a) **Case 1: All 3 higher-level courses from $\{H_1, \dots, H_5\}$** Must include L_1 , so choose remaining 3 lower-level courses from $\{L_2, \dots, L_8\}$: $\binom{7}{3}$. Number of ways to choose higher-level courses: $\binom{5}{3}$. Total for this case: $\binom{7}{3} \cdot \binom{5}{3} = 35 \cdot 10 = 350$.
- (b) **Case 2: All 3 higher-level courses from $\{H_6, \dots, H_{10}\}$** Must include L_2 and L_3 , so choose remaining 2 lower-level courses from $\{L_1, L_4, \dots, L_8\}$ (6 courses): $\binom{6}{2}$. Number of ways to choose higher-level courses: $\binom{5}{3} = 10$. Total: $15 \cdot 10 = 150$.
- (c) **Case 3: Mixed higher-level courses (some from $\{H_1, \dots, H_5\}$ and some from $\{H_6, \dots, H_{10}\}$)**

Subcases for 2+1 or 1+2:

- 2 from $\{H_1, \dots, H_5\}$, 1 from $\{H_6, \dots, H_{10}\}$: Must include L_1, L_2, L_3 (3 lower-level courses fixed). Choose 1 more lower-level course from $\{L_4, \dots, L_8\}$: $\binom{5}{1} = 5$. Number of ways to choose higher-level courses: $\binom{5}{2} \cdot \binom{5}{1} = 10 \cdot 5 = 50$. Total: $5 \cdot 50 = 250$.
- 1 from $\{H_1, \dots, H_5\}$, 2 from $\{H_6, \dots, H_{10}\}$: Must include L_1, L_2, L_3 (3 lower-level courses fixed). Choose 1 more lower-level course from $\{L_4, \dots, L_8\}$: $\binom{5}{1} = 5$. Number of ways to choose higher-level courses: $\binom{5}{1} \cdot \binom{5}{2} = 5 \cdot 10 = 50$. Total: $5 \cdot 50 = 250$.

Total valid curricula with prerequisites:

$$350 + 150 + 250 + 250 = 1000$$

Solution 08: Bank Fraud Monitoring Across Multiple Branches

Let us define events:

- F : Transaction is fraudulent.
- H : Transaction involves a high transfer (\$10,000+).
- I : Transaction is international.

We are given:

$$\begin{aligned}P(F) &= 0.03, & P(F^c) &= 0.97, \\P(H | F) &= 0.40, & P(I | F) &= 0.10, P(H \cap I | F) = 0.05, \\P(H | F^c) &= 0.0005, & P(I | F^c) &= 0.001, P(H \cap I | F^c) = 0.00001.\end{aligned}$$

Part (a): Probability that flagged transaction (both high and international) is fraudulent

We want $P(F | H \cap I)$. By Bayes' theorem:

$$P(F | H \cap I) = \frac{P(H \cap I | F)P(F)}{P(H \cap I | F)P(F) + P(H \cap I | F^c)P(F^c)}.$$

Substitute the values:

$$P(F | H \cap I) = \frac{0.05 \cdot 0.03}{0.05 \cdot 0.03 + 0.00001 \cdot 0.97} = \frac{0.0015}{0.0015 + 0.0000097} \approx \frac{0.0015}{0.0015097} \approx 0.994.$$

Answer: Approximately 99.4% of flagged transactions are fraudulent.

Part (b): Probability that a transaction is fraudulent given that it is either high or international (or both)

The event of interest is $H \cup I$. Using Bayes' theorem:

$$P(F | H \cup I) = \frac{P(H \cup I | F)P(F)}{P(H \cup I | F)P(F) + P(H \cup I | F^c)P(F^c)}.$$

Use inclusion-exclusion:

$$\begin{aligned}P(H \cup I | F) &= P(H | F) + P(I | F) - P(H \cap I | F) \\&= 0.40 + 0.10 - 0.05 = 0.45, \\P(H \cup I | F^c) &= P(H | F^c) + P(I | F^c) - P(H \cap I | F^c) \\&= 0.0005 + 0.001 - 0.00001 = 0.00149.\end{aligned}$$

Now compute:

$$P(F \mid H \cup I) = \frac{0.45 \cdot 0.03}{0.45 \cdot 0.03 + 0.00149 \cdot 0.97} = \frac{0.0135}{0.0135 + 0.0014453} = \frac{0.0135}{0.0149453} \approx 0.903.$$

Answer: Approximately 90.3% of transactions that are high or international are fraudulent.